



Manual Spray Booth for Thermal Arc Spraying Small Parts

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
Manual Spray Booth for Thermal Arc Spraying Small Parts 1				
1.0 Introduction.....2				
2.0 Best Practice Description3				
3.0 Implementation Steps.....3				
4.0 Benefits6				
5.0 Resources Required.....6				
6.0 Supporting Attachments / References6				
7.0 Related Best Practices6				
8.0 Acknowledgements6				

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1.0 Introduction

Information contained in this document is intended for all dealers globally when utilizing thermal spray arc technologies as a primary method for component salvage. This document provides an overview of the tooling requirements for this manual thermal spray cabinet.

This inexpensive system was developed to allow small components (exhaust manifolds, journals, powertrain yokes, etc.). It is a cost-effective solution as it improves the thermal spray capabilities and capacities of the shop with a minimal investment.



Image 1. Manual Spray Booth

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2.0 Best Practice Description

Complex robotic spraying booths are commonly needed for major component salvage (cylinder blocks, spindles, wheels, housings, etc.) MATCO's demand for smaller components exceeded the capacity of the current enclosed spray booth. The addition of a second large spray booth would have required significant capital investment. To optimize the resources of the shop and improve production output the MATCO Engineering team decided to implement a handheld thermal spray booth. This solution eliminated the congestion caused by the smaller components.

3.0 Implementation Steps

3.1 Defining the scope: The two important variables Matco took into consideration when deciding on the capacity of the cabinet, was the size and weight of the components Matco would manually thermal spray weld. They elected to convert an existing grit blast cabinet that was sufficient in size.

3.2 Preparing the cabinet: A rotating table with a variable frequency drive was installed inside the cabinet so that part rotation speed could be made adjustable. The table can include T-slots that would allow a variety of part holding solutions to be used. The cabinet should allow for dust and fumes extraction hoses to be mounted. Allows for flexible positioning of the spray gun.

3.3 Turntable details: The assembly consist of an electric motor, gearbox, mounting surface (Table with T-slots) and a variable frequency drive (VFD). The VFD permits the operator to adjust the desired rotation speed of the table, which is essential when spraying components of varying diameter and shape.



Image 2. Turntable with VFD

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3.4 Work holding: It is necessary to clamp and position a variety of components of varying sizes and shapes. In addition, the part must be held rigid during the process. Matco uses a T-slot clamping kit to allow the operator flexibility in mounting the components to the table. It may be necessary for the end user to develop specialized fixtures or jigs for the various components they will be spraying.

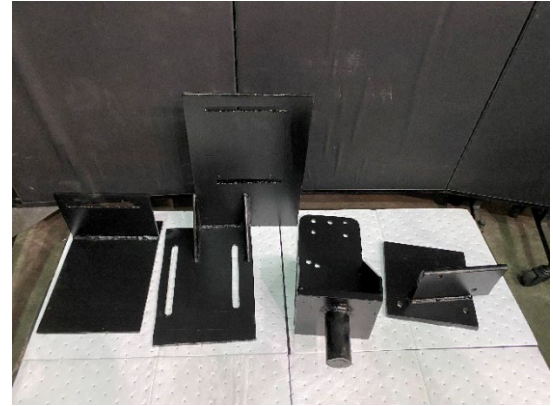


Image 5. T-slot Clamping Kit and Fixtures

3.3 Selecting Thermal Arc spray equipment:

Matco elected to use TAFE 8830 for their application, although other equipment could be selected based on the needs. The power control unit was installed at the rear of the cabinet.



Image 6. Power control unit behind Cabinet



Image 7. Spray Gun

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Manual Spray Booth for Thermal Arc Spraying Small Parts	DATE 11/10/2023	CHG NO 00	NUMBER 4.02-231110-1

3.5 Debris and Fumes Extraction:

Since the arc spray process generates a considerable amount of fumes and debris it was necessary to incorporate a means to extract these from the cabinet. MATCO selected a Donaldson system for this requirement. It was mounted outside and adjacent to the machine shop wall.



Image 8. Debris and fumes extraction unit (As for reference only)

3.6 Spraying Process:

Since the process has been placed in service the following photos show that it's capable of salvaging small components via the arc spray welding process. Control parameters must be maintained to achieve acceptable results. Standoff distance, table rotation speed, voltage, current, temperature, and operator experience can all effect the results. Further technical information on spray welding can be referenced in SEBF2125 Thermal spray procedures for exhaust manifolds.



Image 7. Thermal spray in process



Image 8. Exhaust manifold after process

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CHG
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4.0 Benefits

- 4.1 When properly applied the thermal arc spray process is an excellent process to salvage components and in turn reduce owning and operating cost for the customer.
- 4.2 Minimal floor space requirements.
- 4.3 Versatility for spraying applications.
- 4.4 Low-cost solution. No robot or complex systems required.

5.0 Resources Required

- 5.1 Thermal Spray Equipment (TAFE 8830 or similar)
- 5.2 Turntable with variable frequency drive and gearbox
- 5.3 Debris and Fumes extraction system (Donaldson or similar)
- 5.4 Grit Blast Cabinet (Econoline or similar)

6.0 Supporting Attachments / References

- 6.1 SEBF2125 Thermal Spray Procedures for Exhaust Manifolds
- 6.2 SEBF8048 Procedure to Machine Damaged Transmission and Differential Yokes
- 6.3 SEBF9238 Fundamentals of Arc Spray for Reconditioning Components.

7.0 Related Best Practices

- 7.1 3500 Exhaust Manifold Salvage 2009-0601-1974

8.0 Acknowledgements

This Best practice was submitted by:

Oscar J. De La O / Technical Consultant
 E-mail odelao@matco.com.mx
 Company: Empresas MATCO S.A. de C.V.

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